

Department of Computer Science and Engineering (CSE)
BRAC University



Summer 2024

CSE250 – Circuits and Electronics

Md. Rafatul Haq (Initial: RFH)
Lecturer
Department of CSE
BRAC University
Email: rafatulhaq123@gmail.com

Consultation hours

Saturday	: 12:30 pm – 3:20 pm
Sunday	: 3:30 pm – 5:00 pm
Tuesday	: 3:30 pm – 5:00 pm

Course Outline

- Fundamental electrical concepts and measuring units of **Voltage, Current, Resistance, and Power**
- Laws of Electricity (**Ohm's law, Kirchhoff's Current and Voltage law**) and various methods of electrical circuit analysis (**Nodal, Mesh**)
- Introduction to basic electrical circuit elements
- I-V characteristics
- Circuit analysis in Direct current, First-order Transient, and Alternating current modes, for various combinations of **Resistive, Inductive, and Capacitive networks**
- Phasor representation of sinusoidal quantities; Circuit theorems for linear circuits (**Source Transformation, Superposition, Thevenin, Norton and Maximum Power Transfer**)
- Open loop and closed loop configuration of **Operational Amplifiers**

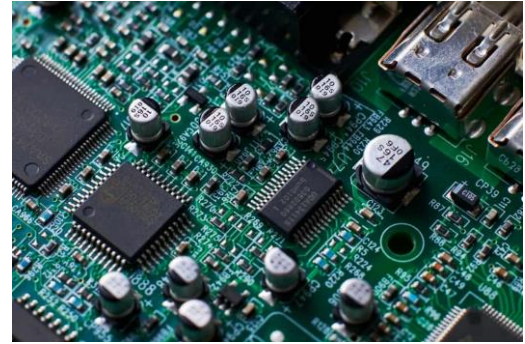
THIS COURSE INCLUDES COMPULSORY 3-HOUR LABORATORY WORK.

Why Should a CSE Student Take this course?

- **Embedded Systems and IoT (Internet of Things)**

Job Market: There is a high demand for engineers who can develop smart devices like home assistants, wearables, and industrial sensors.

How This Course Helps: Knowledge of circuits and electronics is crucial for designing and programming these devices. Understanding hardware allows you to create better software for these systems.



- **Robotics and Automation**

Job Market: Robotics is a growing field with applications in manufacturing, healthcare, and home automation.

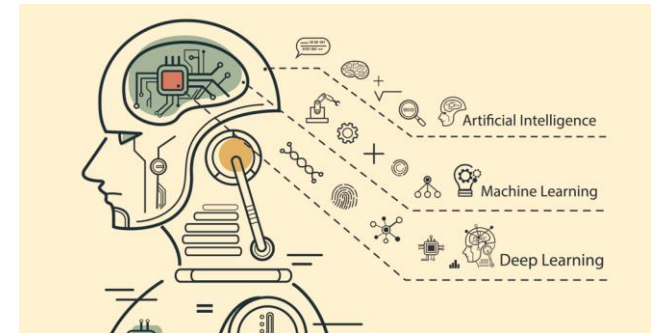
How This Course Helps: Designing and programming robots requires knowledge of both electrical circuits (sensors and motors) and software (control algorithms). This course gives you the foundation to work in this exciting area.



- **AI and Machine Learning Hardware**

Job Market: AI is not just about software; specialized hardware like GPUs and TPUs are crucial for AI applications.

How This Course Helps: Understanding the electronics behind these processing units can help you optimize AI algorithms and develop new AI hardware solutions.



Why Should a CSE Student Take this course?

- **Cybersecurity**

Job Market: Protecting hardware from cyber threats is as important as securing software.

How This Course Helps: Knowledge of electronics can help you understand hardware vulnerabilities and develop more secure systems.



- **Renewable Energy and Smart Grids**

Job Market: There's an increasing focus on sustainable technology and energy-efficient systems.

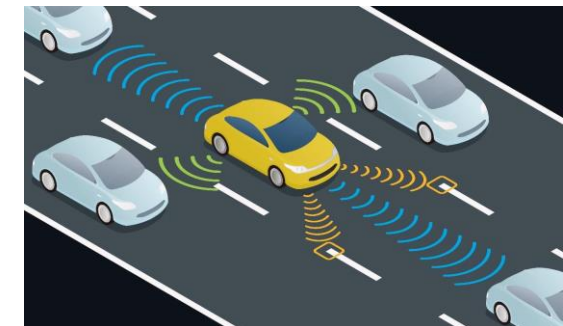
How This Course Helps: EEE knowledge is essential for developing smart grid technology and integrating renewable energy sources with software systems that manage energy distribution.



- **Automotive Industry**

Job Market: Modern cars are filled with electronic systems, from infotainment to advanced driver-assistance systems (ADAS).

How This Course Helps: Knowledge of circuits and electronics is crucial for developing and maintaining these systems, ensuring they work correctly and safely.



Lesson framework

No	Topic	Lecture #
1	Illustrating the motivation behind taking this course. What are the real-life implications of these course materials?	Lecture 1
2	Discuss basic circuit parameters like voltage, current, energy and power definitions and units. Introducing passive sign convention, positive-negative voltage/current/power. Discuss different types of circuit elements (active, passive), and different types of sources (DC/AC, voltage/current, dependent/independent). Introducing circuit symbols.	Lecture 2
3	Introducing basic electrical components: resistors, voltage source, current source. Basic laws of electrical circuits: Ohm's law. I-V characteristics of a resistor. Discuss passive sign convention, finding power of circuit elements by $P=VI$. Defining nodes, loops and mesh. Discuss various circuit configurations: Series, Parallel and others. How to identify series and parallel connections and calculate equivalent resistance. Open and short circuit.	Lecture 3
4	Defining Node/Supernode. Introducing Current Sign Convention. Basic laws of electrical circuits: Kirchhoff's current law. Statement and application of KCL. Current divider rule in a parallel circuit. Illustrating convention doesn't change the KCL equation. Show usefulness of supernode.	Lecture 4

Lesson framework (Continued)

No	Topic	Lecture #
5	Defining Mesh/Supermesh. Revisiting Passive Sign Convention. Basic laws of electrical circuits: Kirchhoff's voltage law. Statement and application of KVL. Voltage divider rule in a series circuit. Illustrating the assumption of the current direction doesn't change the KVL equation. Show usefulness of supermesh.	Lecture 5
6	I-V characteristics of basic circuit elements: Resistor, Voltage source, Current source, Open circuit, Short circuit, any two-terminal device/circuit, a combination of elements (e.g. voltage/current source in series/parallel with resistor). Idea of circuit equivalence. Series-parallel equivalent circuit for resistance/voltage source/current source combinations. Ideal/ non-Ideal current/voltage source. Simplifying circuits by means of equivalence. Basic circuit theorem: Source Transformation theorem. Failure of applying in Wheatstone bridge circuit.	Lecture 6
QUIZ 1		
7	Explaining Nodal Analysis technique, using it to solve for current, voltage, and power in a given circuit (multiple examples).	Lecture 7
8	Reintroducing dependent sources. Demonstrating Nodal Analysis with dependent sources. Problems with floating voltage sources, using Supernodes to solve such circuits.	Lecture 8
9	Explaining MeshAnalysis technique, using it to solve for current, voltage, and power in a given circuit (multiple examples).	Lecture 9

Lesson framework (Continued)

No	Topic	Lecture #
10	Demonstrating Mesh Analysis with dependent sources. Problems with common current sources, using Supermeshes to solve such circuits.	Lecture 10
11	Linear circuit elements. Linearity of voltage, current in circuits, and non-linearity of power. Circuit theorem: Superposition theorem. Using superposition theorem for solving DC circuits. Superposition Theorem for circuits with Dependent Sources.	Lecture 11
QUIZ 2		
MIDTERM		
12	Reintroduction to circuit linearity, I-V characteristics of linear circuits. Circuit Theorems: Thevenin's theorem. The motivation behind Thevenin's theorem.	Lecture 12
13	Using test voltage/current sources while deactivating sources to find Thevenin's. Solving resistance matching problems for transferring maximum power. Norton's theorem, the relation between Thevenin's and Norton's theorem.	Lecture 13
14	Using Thevenin's/Norton's theorem for solving circuits. Maximum transferable power and condition for it.	Lecture 14
QUIZ 3		

Lesson framework (Continued)

No	Topic	Lecture #
15	Capacitors and Inductors, their component equations. SI unit for measuring capacitance and inductance. Transient circuits, visualizing and analyzing transient circuits.	Lecture 15
16	Response of transient circuit: first order RC circuit, time constant. Analyzing and plotting first-order transient circuit response. Finding capacitor current from capacitor voltage	Lecture 16
17	Response of transient circuit: first order RL circuit, time constant. Analyzing and plotting first-order transient circuit response. Finding inductor voltage from inductor current.	Lecture 17
18	Complex number review. Alternating current, the importance of AC circuit. Visualizing the dynamics of an AC circuit, Amplitude, RMS voltage/current and finding them from a graph.	Lecture 18
19	Introducing Impedance. Defining impedance for various elements, Phasor analysis of an AC circuit. Instantaneous voltage, current and power. Applying superposition theorem on AC circuits containing sources of different frequencies	Lecture 19
QUIZ 4		
FINAL EXAM		

Course Materials

Text and reference books

Sl.	Title	Author(s)	Publication Year	Edition	Publisher	ISBN
1	Fundamentals of Electric Circuits	Charles K. Alexander, Matthew N. O. Sadiku	2019	6th	McGraw Hill Education	978-9353165505
2	Introductory Circuit Analysis	Robert L. Boylestad	2013	12th	Pearson Education India	978-9332518612
3	Electric Circuits	James W. Nilsson Susan A. Riedel	2010	9th	Pearson College Div	978-0136114994

Other materials (Link)

[Lecture Slides](#)

[Video Lectures](#)

[Practice Problems](#)

[Suggested Problems](#)

Marks Distribution

Assessment	Criteria	Weightage (%)
Attendance		10
Assignment	2 to 4	5
Quiz	Best 3 out of 4	15
Midterm	Central Exam	25
Final	Central Exam	25
Lab	Hardware & Software Experiments	20
Total		100



Presence	Grade (out of 10)
80% or above	10
75% to below 80%	9
70% to below 75%	8
65% to below 70%	7
60% to below 65%	6
...	...
Below 50%	0

Thank You